

Real Estate Analytics Report

1. Executive Summary

This project applies data analytics to real estate datasets to uncover pricing trends, demand drivers, and investment opportunities. Using Python, Power BI, and statistical modeling, the study provides actionable insights for buyers, sellers, and investors.

2. Project Overview

The project integrates **data preparation**, **predictive modeling**, and **dashboarding** to deliver a comprehensive view of the property market. It combines machine learning models with interactive dashboards for decision-making.

3. Business Objectives

- Identify key drivers of property prices
- Segment markets into luxury, affordable, and mid-range categories
- Provide dashboards for stakeholders to monitor KPIs
- Recommend strategies for investors and developers

4. Dataset Description

- **Attributes:** Location, property type, square footage, price, amenities
- **Sources:** Property listings and transaction records
- **Size:** Thousands of records across multiple cities

5. Data Preparation

- Cleaning missing values
- Encoding categorical variables (property type, location)
- Normalizing numerical features (price, square footage)
- Outlier detection for extreme property values

6. Data Model

- **Fact Table:** Transactions (price, date, property ID)
- **Dimension Tables:** Property details, location, amenities
- Relationships modeled in Power BI for efficient querying

7. DAX & KPIs

- **Average Price** = $\text{SUM}(\text{Price}) / \text{COUNT}(\text{Properties})$

- **Price per Sqft** = $\text{SUM}(\text{Price})/\text{SUM}(\text{Sqft})$
- **Growth %** = $(\text{Current Period} - \text{Previous Period})/\text{Previous Period}$
- **Market Share by Property Type**

8. Property Market Analysis

- Urban centers show higher volatility
- Suburban markets are more stable
- Luxury properties dominate in prime locations

9. Price Analysis

- Regression models achieved $R^2 \sim 0.65$
- Random Forest improved accuracy to $R^2 \sim 0.78$
- Location and property size emerged as strongest predictors

10. Location Analysis

- Heatmaps reveal demand hotspots in city centers
- Suburban areas show steady growth
- Proximity to schools and transport hubs increases property value

11. Property Type Analysis

- Apartments: Most common, moderate pricing
- Villas: High-value, concentrated in luxury clusters
- Plots: Variable pricing, strong investment potential

12. Key Dashboard Insights

- KPI cards highlight average price and growth trends
- Clustered bar charts compare property types
- Interactive filters allow drill-down by location and amenities

13. Business Recommendations

- Target mid-range properties in emerging suburbs for stable returns
- Developers should prioritize amenities like parking and school access
- Monitor urban volatility for short-term investment opportunities

14. Conclusion

The project demonstrates how analytics can guide pricing strategies, investment decisions, and market positioning. Future enhancements include **time-series forecasting**, **geospatial analysis**, and **interactive dashboards** for real-time insights.